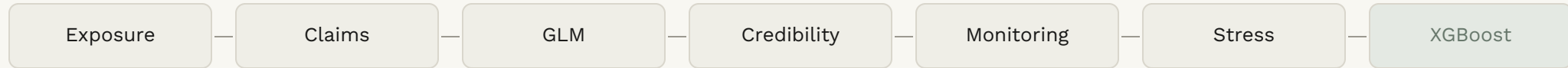


Return-Shipping Insurance Pricing Engine for E-commerce

Exposure-level actuarial pricing workflow with GLM, seller credibility, monitoring, stress testing, and XGBoost challenger.



The product prices return-shipping loss, not generic refund behavior.

Coverage A

Return Shipping

Covered loss = compliant buyer return shipping cost.

Explicitly not covered

- Product refund itself
- Fraud or refund without return
- Failed delivery / replacement shipping
- Non-delivered exposure

A return event is not automatically an insurance claim. The claim layer only pays covered return-shipping cost.

Correct exposure definition is the foundation of pricing.

order_id + order_item_id + seller_id + product_id

Total rows

112,650

Eligible exposures

110,197

Ineligible exposures

2,453

Portfolio

Seller

Order

Order-item exposure

Why not order_id alone? One order can contain multiple products and sellers, while freight_value is recorded at item level.

Olist has real commerce logistics data,
but no insurance claim history.

Returns layer

return_requested, return_approved,
refund_without_return, return_reason

Claims layer

covered_claim_flag, claim_type,
gross_loss, paid_loss, net_loss

Eligible exposures

110,197

Covered claims

8,485

Avg severity

19.22

Frequency

7.70%

Total net loss

163,112.68

Frequency target = covered_claim_flag. Severity target = net_loss on covered claims only.

Start with a simple actuarial benchmark before modelling.

Portfolio pure premium

total net loss / eligible exposures

Commercial premium

pure premium / target loss ratio

Target loss ratio

60.00%

Portfolio pure premium

1.4802

Commercial premium

2.4670

Baseline is the control model: everyone gets the same price, so it cannot differentiate category, route, freight, or seller risk.

GLM is the current champion because it is explainable and calibrated.

Frequency

Binomial GLM
covered_claim_flag

Severity

Gamma GLM
net_loss on covered claims

Expected loss = frequency x severity

Metric	Test result
Frequency AUC	0.560
Pure premium A/E	1.003
Predicted pure premium	1.503
Actual pure premium	1.508

Feature rule: quote-time-known only. Outcome, post-bind, and synthetic generation fields are excluded.

Seller experience matters, but direct seller_id modelling is unstable.

Credibility formula

$$Z = n / (n + 500)$$

$$\text{seller relativity} = Z \times \text{observed A/E} + (1 - Z) \times 1.0$$

Shrink target = seller's GLM base expected loss,
not portfolio average price.

Sellers

2,970

Median exposures

8

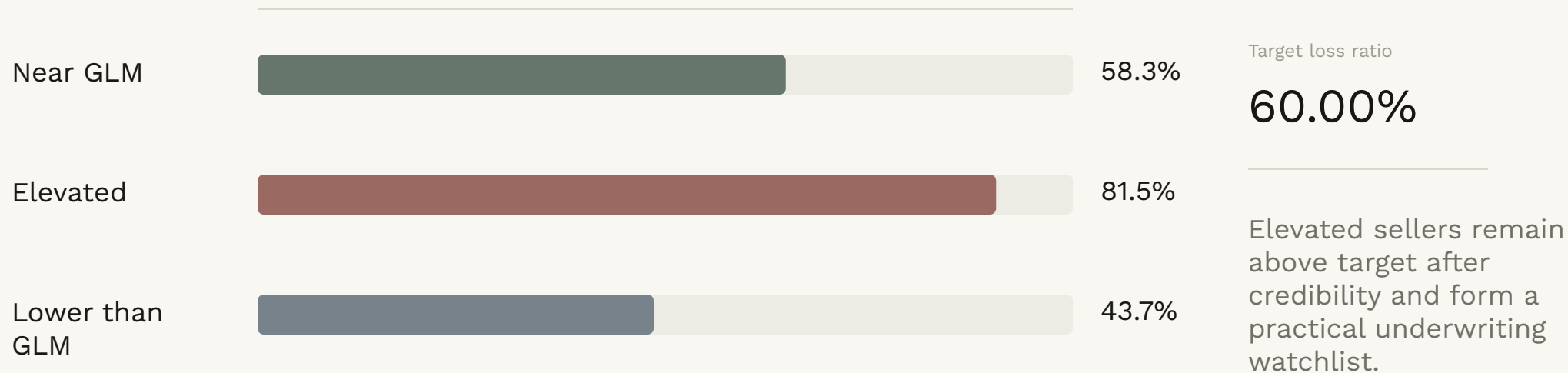
Sellers >= 500 exposures

29

Portfolio normalization preserves total GLM expected loss after seller relativities.

This lets large sellers carry more of their own experience while small sellers remain close to their GLM risk mix.

Portfolio A/E close to 1 is not enough; monitor by segment.



Portfolio GLM + credibility loss ratio = 59.37%

A calibrated model can still be fragile under adverse scenarios.



Base conditions are close to target, but simultaneous frequency and severity deterioration creates high pressure.

XGBoost is tested as challenger, not assumed to be the final rate basis.

Parameter sweep insight

First run: train AUC 0.635, test AUC 0.555
Regularized: train AUC 0.589, test AUC 0.561

Test metric	GLM	XGBoost
Frequency AUC	0.5598	0.5609
Pure premium A/E	1.0030	1.0145
Predicted pure premium	1.503	1.486
Loss ratio	60.18%	60.87%

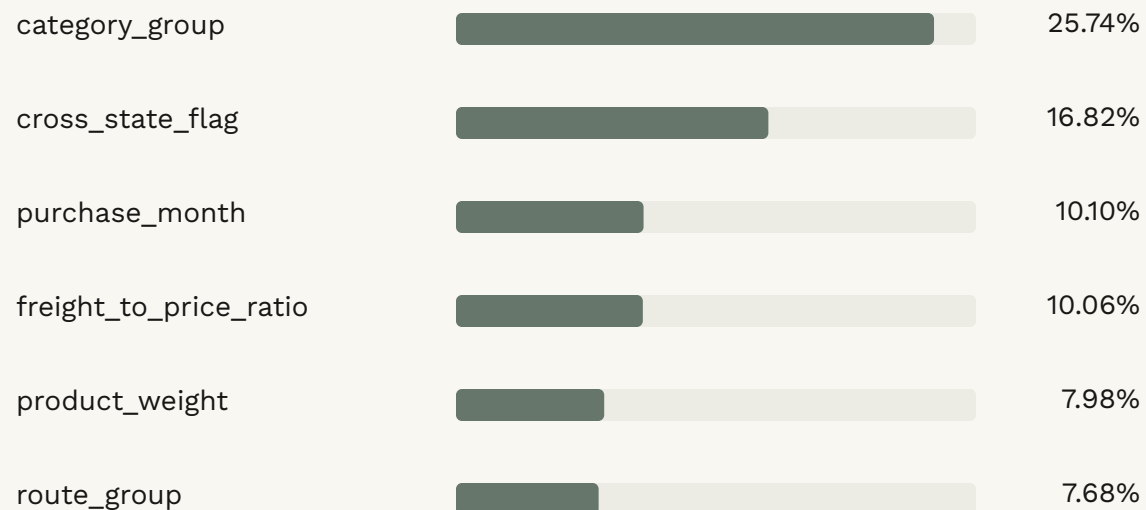
Conclusion

Slightly better frequency ranking, but weaker pure premium A/E and loss-ratio calibration.

Therefore XGBoost remains a challenger risk score and monitoring benchmark, not the champion pricing formula.

The challenger uses plausible signals, but interpretation does not override calibration.

Frequency: top base features by native SHAP-style mean absolute contribution



Severity

freight_value_capped

80.69% contribution share

Method note: these are XGBoost native contribution summaries using `pred_contribs=True`, not external shap package visualizations.

Keep GLM + seller credibility as champion pricing.

Use XGBoost as challenger risk score and monitoring layer.

Why

GLM is explainable and better calibrated. Credibility handles seller experience in a controlled actuarial way.

Do not overclaim

XGBoost adds a nonlinear benchmark but does not yet justify production replacement.

Production roadmap

Real order / return /
claim data

Historical train + future
validation

Claim development /
IBNR

Expense, commission,
capital loads

Underwriting rules for
elevated sellers

Drift, A/E, LR, stress
monitoring

The value is not one model score. The value is a pricing workflow connecting exposure definition, expected loss, credibility, monitoring, stress testing, and governance.